



Date	23 July 2026
Project Name	SB-STOCKS
Maximum Marks	2

Solution Architecture

Architecture Overview

MERN Stack - 3-tier architecture (Client, Server, Database)

Architecture Components

Layer	Technology	Description
Presentation	React.js + Bootstrap	UI components, dashboards, trade modals
Application	Express.js + Node.js	REST API, business logic, authentication
Data	MongoDB + Mongoose	Data persistence, schema validation
Security	JWT + bcryptjs	Token-based auth, password encryption

Architecture Diagram

(Diagram space reserved for Solution Architecture diagram)

Data Flow: The user interacts with the React interface. API requests are made via Axios to the Node/Express backend. The backend validates JWT tokens, processes business logic, and performs CRUD operations with the MongoDB database using Mongoose.